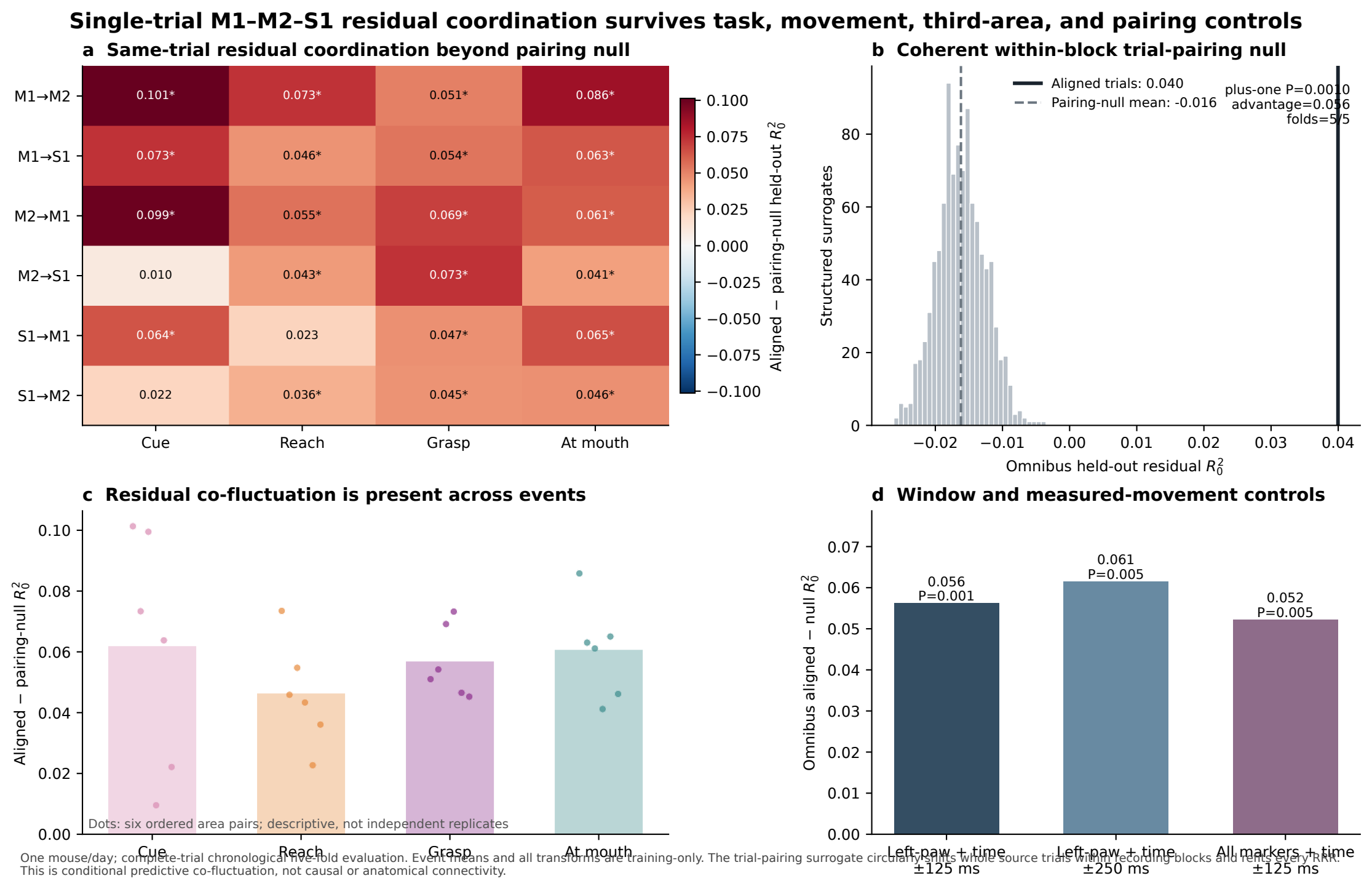
Single-trial M1–M2–S1 residual coordination survives task, movement, third-area, and pairing controls



One mouse/day; complete-trial chronological five-fold evaluation. Event means and all transforms are training-only. The trial-pairing surrogate circularly shifts whole source trials within recording blocks and refits every RRR. This is conditional predictive co-fluctuation, not causal or anatomical connectivity.